



Method

- Each block computes $\mathbf{y} = \mathbf{F}(\mathbf{x}, \{\mathbf{W}_i\}) + \mathbf{x}$ — the **2–3-layer residual $\mathbf{F}$** is added through an identity shortcut that introduces zero extra parameters and almost no extra compute over a plain stack.
- When channel count or spatial size changes, the shortcut either **zero-pads** (option A) or applies a **1×1 projection** (B); deeper nets stack **1×1, 3×3, 1×1 bottleneck blocks** for compute efficiency at extreme depth.
- BatchNorm after every conv, **He initialization**, SGD batch 256, LR 0.1 with step decay, weight decay 1e–4, momentum 0.9, **no dropout** ; 60×10⁴ iterations on ImageNet.
- Downsampling lives in stride-2 convs at **conv3_1 / conv4_1 / conv5_1**; CIFAR-10 ResNets follow a **6n+2** recipe with identity shortcuts everywhere.

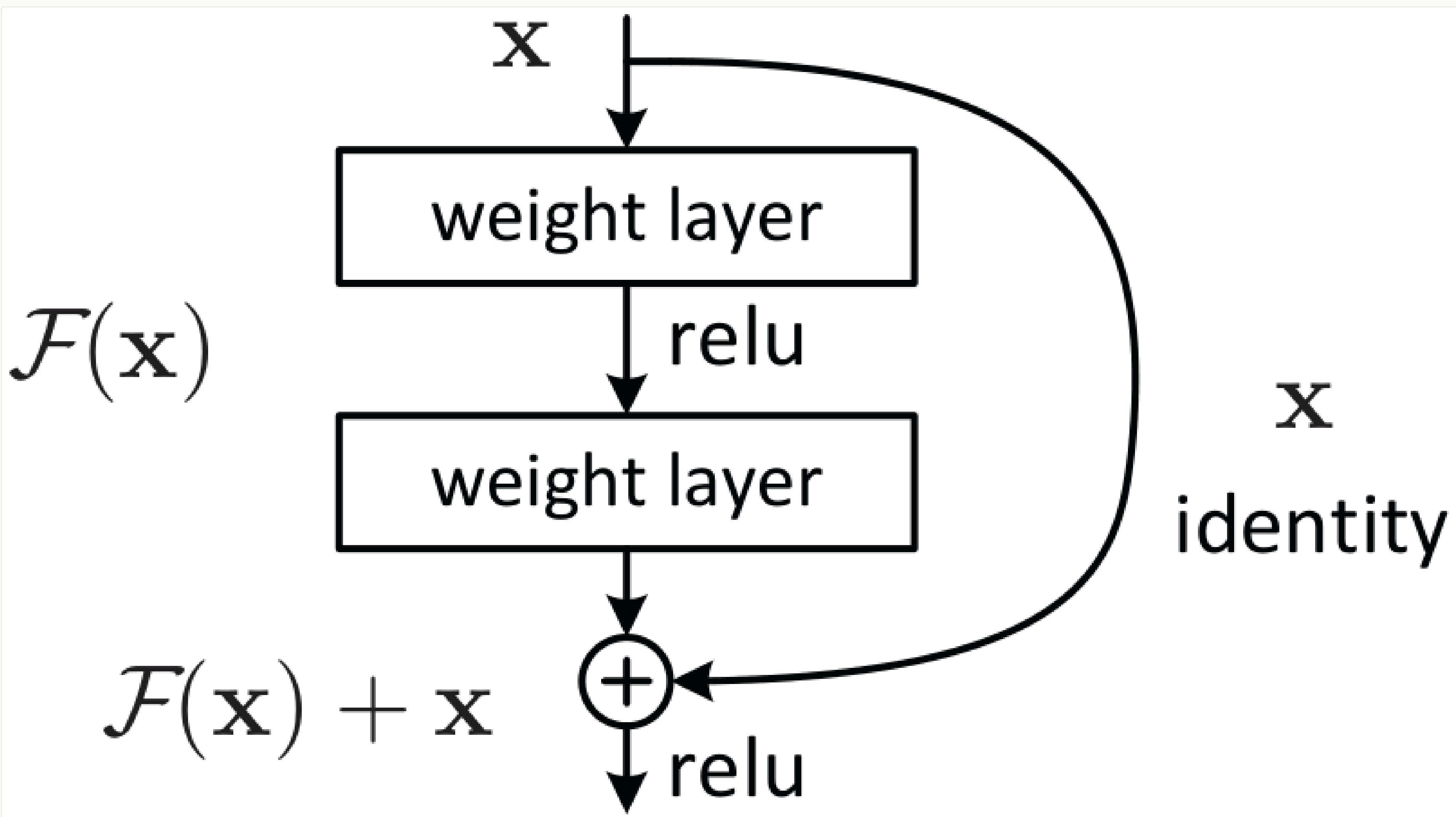


Fig. 2 · A residual building block: two weight layers learn $\mathbf{F}(\mathbf{x})$, and an identity shortcut adds $\mathbf{x}$ back — the entire framework adds no extra parameters and almost no extra compute.

Problem

Plain deep networks suffer a **degradation problem**: as depth grows past a point, training accuracy **saturates and then gets worse** — not from overfitting, but from optimization difficulty. A 56-layer plain CNN has **higher** training error than a 20-layer one on CIFAR-10: deeper actively hurts.

Motivation

- BatchNorm + He init** already tame vanishing gradients — gradient norms stay healthy at 30+ layers, so degradation is NOT a gradient-flow issue.
- A deeper net can match a shallower one by setting extra layers to identity — degradation is a **pure optimization failure** of SGD on very deep stacks, not a representational limit.

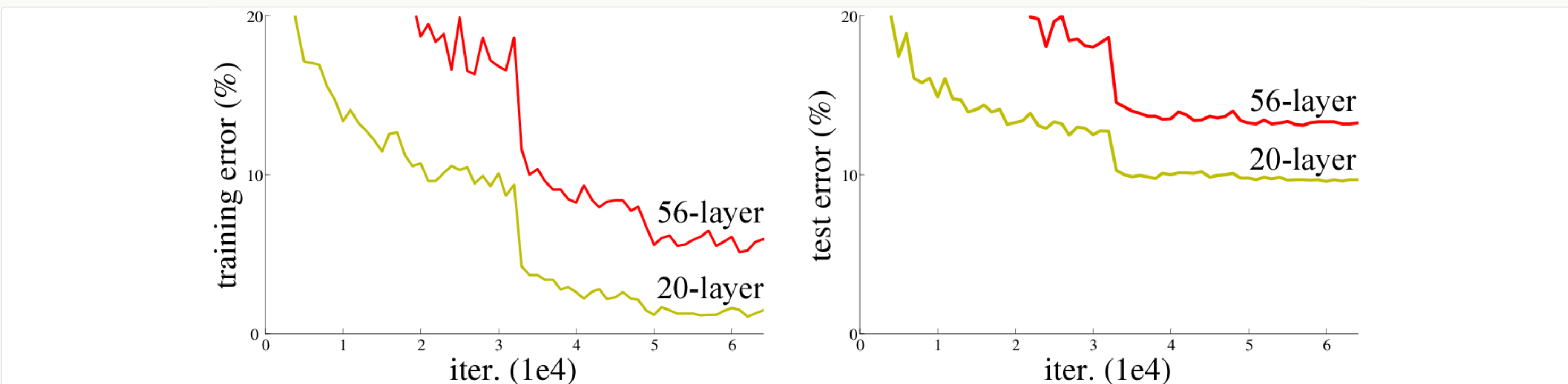


Fig. 1 · CIFAR-10 train (left) / test (right): the 56-layer plain net loses to the 20-layer one at every step. Both train AND test error rise — ruling out overfitting and exposing pure optimization failure.

Takeaway

Residual learning with identity shortcuts **removes the degradation barrier** — accuracy keeps improving as depth grows into the hundreds, turning **depth from a liability into an asset**. The same ResNet backbones swept the **2015 ILSVRC + COCO challenges**, and the residual block became the default primitive across vision — reused in Transformers, U-Nets, diffusion.

Key Results

Method	Top-1	Top-5
Plain-34 (baseline)	28.54	10.02
ResNet-34	25.03	7.76
ResNet-152 (single)	21.43	4.49
ResNet-152 ×6 (ensemble)	—	3.57

–**3.51 pts top-1** from Plain-34 to ResNet-34 — the degradation curve flips: ResNet-34 BEATS ResNet-18 by 2.8 pts top-1, the opposite of the plain-net trend.

Same ResNet-101 backbone, dropped into Faster R-CNN, lifts COCO mAP@[.5,.95] by **+28% relative** over VGG-16. The 152-layer net is **8× deeper** than VGG-19 with lower theoretical FLOPs.

Ablation Study

- Shortcut type** (Tbl. 3, ResNet-34): zero-pad **A 25.03 / 7.76** → proj-on-change **B 24.52 / 7.46** → everywhere **C 24.19 / 7.40** — small gaps confirm parameter-free identity shortcuts are sufficient.
- Layer-response std** (Fig. 7) drops with depth — deeper ResNets settle to smaller per-layer outputs, so **residuals stay closer to zero**, confirming the framework's central learned-residual hypothesis.

Headline Numbers

3.57%

Top-5 error · ImageNet 2015 (1st, ILSVRC)

152

layers (8× VGG-19)

11.3B

FLOPs (VGG-19: 19.6B)

+28%

rel. COCO mAP